



PARSHVANATH CHARITABLE TRUST'S

A. P. SHAH INSTITUTE OF TECHNOLOGY

Department of Information Technology

(NBA Accredited)



Academic Year: 2026-27

Class/ Branch: BE IT C

Subject: Recent Open Source Lab

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EXPERIMENT NO. 01

Aim: To understand different steps for contributing to open-source projects

Theory:

The open-source community provides a great opportunity for aspiring programmers to distinguish themselves; and by contributing to various projects, developers can improve their skills and get inspiration and support from like-minded people. But most importantly, they can prove that they can build fantastic experiences that people love.

Reason to contribute to open-source projects

1. There are a lot of enthusiasts who simply believe that code should be open. They're idealists who want to make the world a better place, and it drives them to contribute code. The desire to share can be a powerful motivator.
2. OSS gives you a great start. Beginners might start by fixing minor things, such as a bug in a library, sending a pull request, or even writing a piece of documentation. However, beginner developers can also learn to write so-called "clean code" – code that is readable and maintainable – while contributing to open source projects. When developers realize that their code is exposed to the world, it makes them focus on making that code easy to understand and support.
3. Programmers stick to generally accepted rules within a team, which include norms for indents, descriptions of methods and classes, variable names, and following the don't-repeat-yourself rule. In a nutshell, when contributing to free projects you're obliged to conform to the norms of a project.
4. You get the chance to be part of an active open source community where you can meet like-minded people and supporters. Moreover, if you're a freelancer and actively contribute to open source projects, you increase your chances of being noticed by potential employers.
5. The main reasons why developers go for free-for-modification projects are to be recognized, to sharpen their programming skills, and to become part of the vibrant community.



Things to consider while selecting Open Source:

Programming language:

The most fundamental technology behind any application is a programming language. The most popular languages on GitHub (a collaborative code hosting platform) are JavaScript, Python, Java, Ruby, and PHP.

Type of project:

After you've chosen the language, you want to work in, you need to choose the type of project you prefer.

How to contribute to open source projects :

GitHub is the most popular platform for open-source collaboration, so you'll probably use it when exploring the world of OSS.

First, you need to create a GitHub account and read the guide that helps you get started. On GitHub, you can contribute to projects by submitting issues and contributing code.

Submitting issues means sending messages about errors in applications and suggesting ways to fix them.

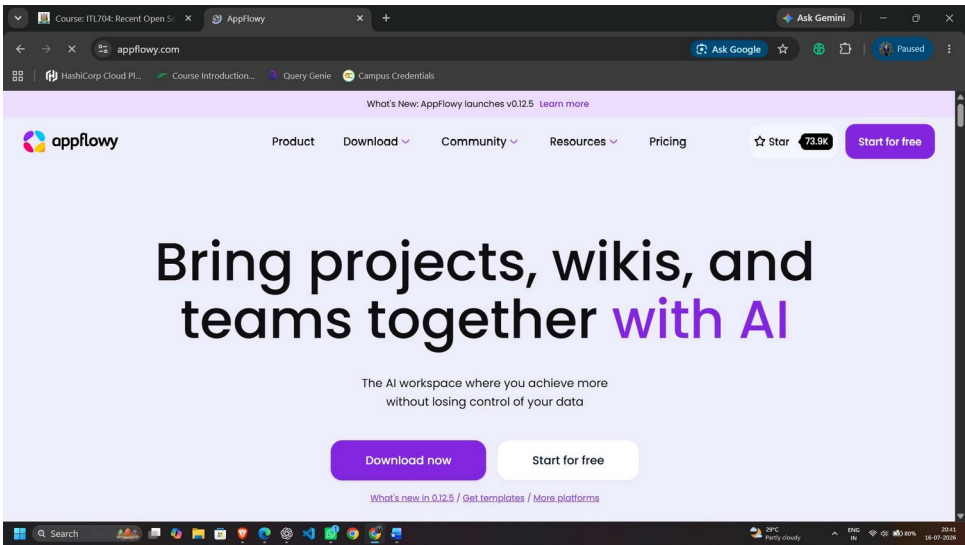
Contributing code involves sending pull requests with your corrections and improvements.

2. Create open source alternatives to commercial software

Today's commercial projects actively engage open source solutions. Many companies base their projects on free tools. When there's a huge selection of software, you don't need to reinvent the wheel. This is why it's useful to play around with free software that can replace similar proprietary software, or that fixes an issue you've recently faced.

Another reason for replacing commercial solutions with open source software is eagerness for real innovation and growth. Commercial software claims to be innovative, but its final goal is turning a profit. Open source software unites best practices, great quality of code and passionate developers willing to code just because they like to.

1. AppFlowy (Alternative to Notion)



Category	Notes & Workspace
Commercial Product	Notion
Open Source Alternative	AppFlowy
GitHub	https://github.com/AppFlowy-IO/AppFlowy

About Notion

Notion is an all-in-one productivity workspace used for note-taking, project management, documentation, task tracking, and collaboration.

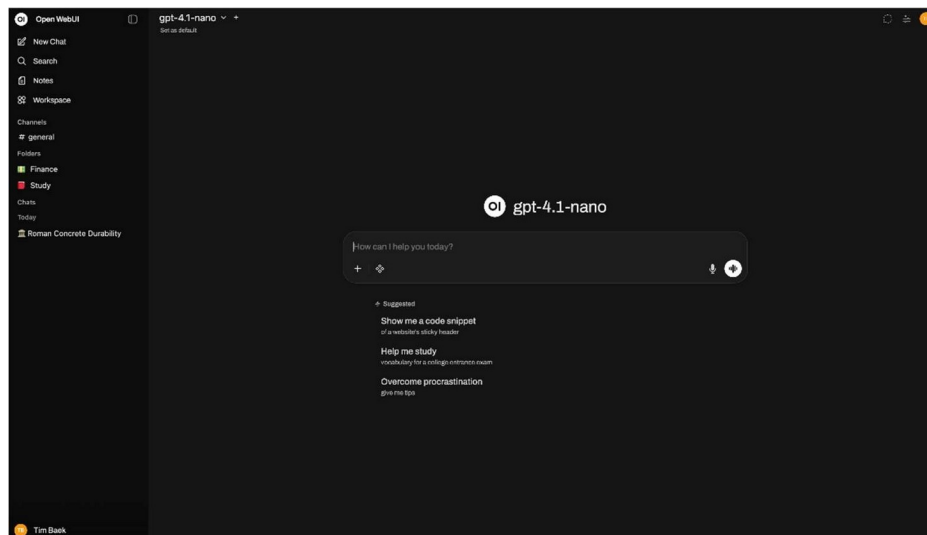
About AppFlowy

AppFlowy is an open-source productivity platform that provides a Notion-like workspace while allowing users to own and control their data.

Common Features

Feature	Notion	AppFlowy
Rich Text Editor		
Notes & Documents		
Kanban Boards		
To-do Lists		
Tables & Databases		
Nested Pages		
Templates		

2. Open WebUI (Alternative to ChatGPT)



Category	AI Assistant
Commercial Product	ChatGPT
Open Source Alternative	Open WebUI + Ollama
GitHub	Open WebUI: https://github.com/open-webui/open-webui Ollama: https://github.com/ollama/ollama

About ChatGPT

ChatGPT is an AI-powered conversational assistant developed by OpenAI. It helps users with answering questions, generating content, writing code, summarizing documents, translating languages, brainstorming ideas, and much more through natural language conversations.

About Open WebUI + Ollama

Open WebUI is an open-source, self-hosted AI chat interface that provides a ChatGPT-like experience. Ollama allows users to run large language models (LLMs) such as Llama, Mistral, Gemma, and others locally on their computer without requiring cloud services.

Common Features

Feature	ChatGPT	Open WebUI + Ollama
AI Chat Interface		
Multi-turn Conversations		
Code Generation		
Document Summarization		

Feature	ChatGPT	Open WebUI + Ollama
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File Upload Support

Markdown Rendering

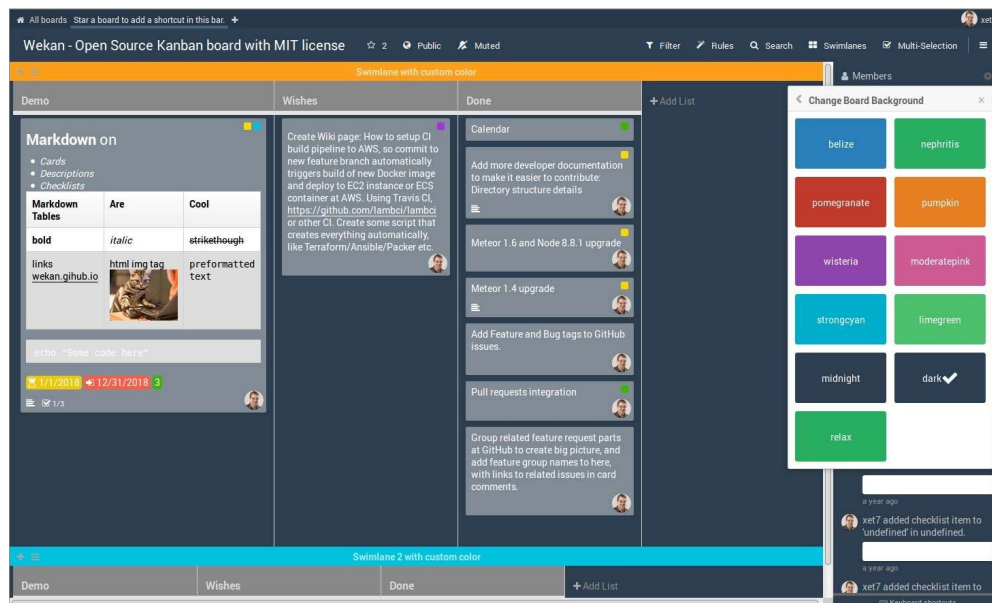
Chat History

Multiple AI Models

Custom AI Prompts

Dark Mode

3. Wekan (Alternative to Trello)



Category	Task Management
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Commercial Product	Trello
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Open Source Alternative	Wekan
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GitHub	https://github.com/wekan/wekan
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About Trello

Trello is a popular project and task management application that organizes work using Kanban boards. It helps individuals and teams manage projects through boards, lists, and task cards.

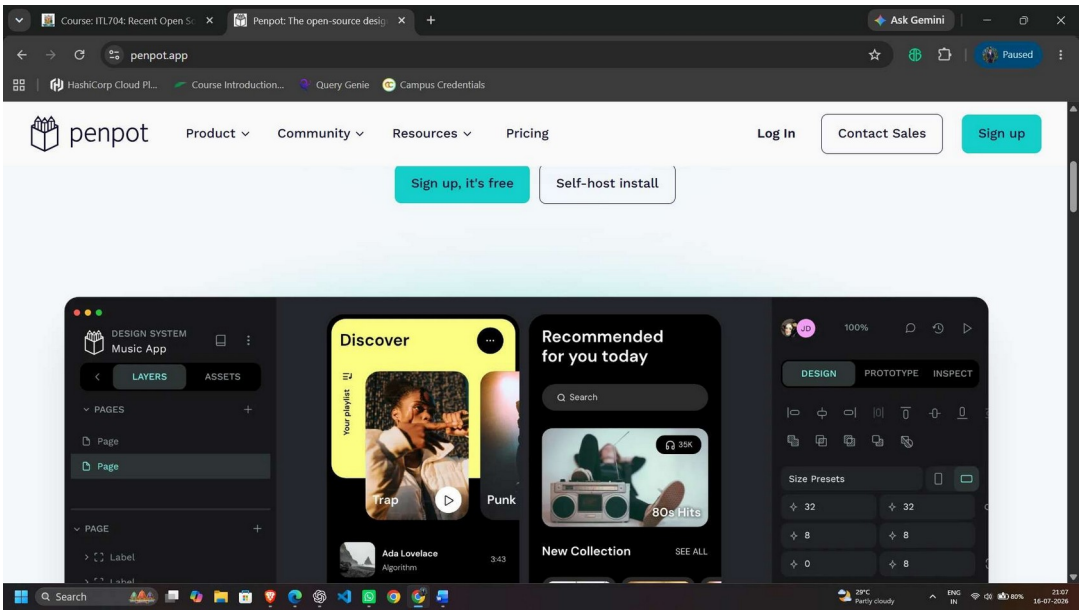
About Wekan

Wekan is an open-source Kanban board application that offers functionality similar to Trello. It enables teams to organize tasks, collaborate on projects, and track progress while allowing complete self-hosting and customization.

Common Features

Feature	Trello	Wekan
Kanban Boards		
Task Cards		
Drag & Drop		
Lists & Columns		
Checklists		
Labels & Tags		
Due Dates		
Team Collaboration		
File Attachments		
Activity History		

4. Penpot (Figma Alternative)



Category	UI/UX Design
Commercial Product	Figma
Open Source Alternative	Penpot
GitHub	https://github.com/penpot/penpot

About Figma

Figma is a cloud-based UI/UX design platform used for wireframing, interface design, prototyping, and real-time collaboration among designers and developers.

About Penpot

Penpot is an open-source design and prototyping platform that offers a Figma-like experience. It enables teams to create UI/UX designs, interactive prototypes, and collaborate in real time while allowing self-hosting.

Common Features

Feature	Figma	Penpot
UI Design		
Vector Editing		
Interactive Prototypes		
Components		
Design Libraries		

Feature	Figma	Penpot
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Team Collaboration

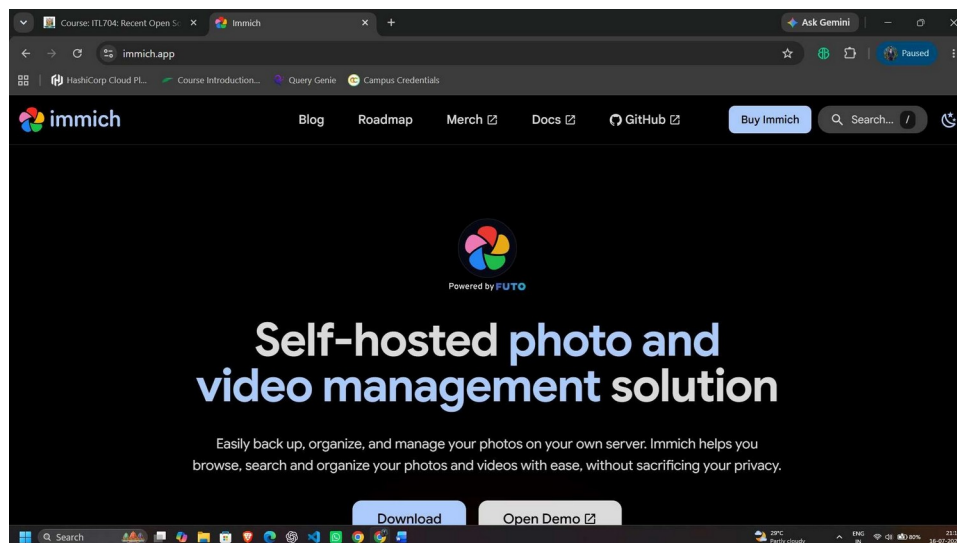
Comments

Version History

Auto Layout

Cross-platform Access

5. Immich (Google Photos Alternative)



Category	Photo Management & Backup
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Commercial Product	Google Photos
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Open Source Alternative	Immich
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GitHub	https://github.com/immich-app/immich
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About Google Photos

Google Photos is a cloud-based photo and video management service developed by Google. It allows users to securely back up, organize, search, edit, and share their photos and videos across multiple devices using AI-powered features.

About Immich

Immich is a free and open-source, self-hosted photo and video backup platform. It provides

functionality similar to Google Photos, enabling users to upload, organize, search, and manage their media while maintaining complete control over their data and privacy.

Common Features

Feature	Google Photos	Immich
Photo Upload		
Video Upload		
Automatic Backup		
Albums		
Search Photos		
Face Recognition		
Map View		
Favorites		
Shared Albums		
Mobile Applications		
Cross-platform Access		
Dark Mode		
Photo & Video Organization		
Metadata Display		
Multi-user Support		

Conclusion

We've listed common reasons why people contribute to OSS projects, and described various ways to get started.